The Non-Hermitian SSH Chain

Skin effect, biorthogonal bulk–boundary correspondence,
and when `eig` stops being trustworthy

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Abstract

We reproduce and extend the biorthogonal bulk–boundary correspondence of Kunst *et al.* [1] for the SSH chain with non-reciprocal intracell hopping. In this model the open-chain edge states appear and disappear far from where the Bloch bands close, and every bulk eigenvector piles up at one end (the non-Hermitian skin effect). We construct an exact, site-resolved similarity map of the open chain onto a Hermitian SSH chain with hopping $v' = \sqrt{t_1^2 - \gamma^2/4}$; the open spectrum agrees with it to 10^{-14}, and the map predicts the skin profile, the open-boundary transition $t_1 = \sqrt{t_2^2 + \gamma^2/4}$ (located from a parameter-free finite-size crossing to four digits) and a second transition for $\gamma > 2t_2$. We implement the biorthogonal polarisation P, derive its leading finite-size term $1 - P = 1/[N(1-x)]$, and show that P is undefined on even chains exactly at the second transition. Finally, using the exact solution as reference, we show that LAPACK's eigensolver stays accurate to 10^{-13} even at eigenvector condition numbers of 10^{24}, and fails abruptly at 76 sites, where `dhseqr` switches to multishift QR, not at $|s|^N \sim 10^{16}$ as a condition-number bound suggests. 60-digit arithmetic confirms the exact formula to 10^{-15}.

1 Introduction

Non-Hermitian Hamiltonians describe open systems with gain, loss or non-reciprocity: photonic lattices, electric circuits, mechanical metamaterials [3]. Their topology differs from the Hermitian case in a striking way. The conventional bulk–boundary correspondence (edge states appear where the Bloch bands close) can fail completely, because under open boundary conditions (OBC) all eigenstates may accumulate at one edge, the *skin effect* [5, 6]. Kunst *et al.* [1] restored a correspondence using biorthogonal quantum mechanics [4]: densities built from left *and* right eigenvectors. Yao and Wang [2] restored it with a generalised Brillouin zone (see the companion non-Bloch project).

Numerically, non-Hermitian matrices are treacherous: eigenvalues of non-normal matrices can be exponentially sensitive to rounding [7]. A study that relies on a general eigensolver must therefore establish *independently* when its output can be trusted. Our approach is to find an exact solution first and use it as the reference for every numerical result.

Contributions.

1. A site-resolved similarity map to a Hermitian SSH chain, used as exact reference. Scaling whole cells uniformly, as is sometimes suggested, leaves the intracell bond non-reciprocal.
2. A parameter-free finite-size locator of the OBC transition, and the second transition for $\gamma > 2t_2$.
3. The biorthogonal polarisation with its derived $1/N$ term, and the even-chain singularity at $x = -1$.
4. A diagnosis of where and why LAPACK's general eigensolver fails on these graded matrices.

Table 1: Summary of the main results ($\gamma = 4/3$, $t_2 = 1$ unless stated).

Quantity	Computed	Analytic
Open-chain eigenvalues vs Hermitian SSH(v', t_2), $N \le 20$	agree to $\le 10^{-14}$	identical
Open spectrum at $t_1 = 0.7$	$\mathrm{Im}\, E = 0$ exactly	real
Skin decay per cell, $t_1 = 0.7, 1.0, 1.4$ ($N = 12$)	0.1591, 0.4508, 0.5959	$1/\lvert s\rvert = 0.1562$, 0.4472, 0.5957
Slope of $\ln \min \lvert E\rvert$, $t_1 = 1.0$	-0.29390	$\ln(v'/t_2) = -0.29389$
OBC transition, $\gamma = 0.5$, 4/3, 2, 2.6	1.0308, 1.2019, 1.4142, 1.6401	$\sqrt{t_2^2 + \gamma^2/4}$ (4 digits)
Biorthogonal edge density vs Hermitian density	$\le 3 \times 10^{-12}$	equal
P from `eig` vs closed form, $N = 11, 21, 31$	7×10^{-10}	equal
P at Bloch points $t_1 = 1/3, 5/3$, $N = 1001$	0.9993, 0.0007	no jump
Finite-size form of P	to 10^{-9}	$1 - P = 1/(N(1-x))$
Second transition, $\gamma = 3$	$t_1 = 1.118$	$\sqrt{\gamma^2/4 - t_2^2}$
Largest N with `eig` error $< 10^{-10}$, $t_1 = 0.7$	$N = 37$ (74 sites)	—
60-digit arithmetic, $N = 40$, $t_1 = 0.7$	exact to $\le 10^{-15}$; `eig` off by 3.7×10^{-5}	—

2 Theory and derivations

2.1 Biorthogonal quantum mechanics

For a non-Hermitian H the right and left eigenvectors are different:

$$H|R_n\rangle = E_n|R_n\rangle, \qquad H^\dagger|L_n\rangle = E_n^*|L_n\rangle. \tag{1}$$

Evaluating $\langle L_m|H|R_n\rangle$ in two ways gives $(E_n - E_m)\langle L_m|R_n\rangle = 0$, so left and right vectors of different eigenvalues are orthogonal. For a diagonalisable H they can be normalised to

$$\langle L_m|R_n\rangle = \delta_{mn}, \qquad \sum_n |R_n\rangle\langle L_n| = \mathbb{1}. \tag{2}$$

This fixes only the product of the two normalisations; we also set $\|R_n\| = 1$. The natural "density" of state n on site i is then the biorthogonal one, $\langle L_n|\Pi_i|R_n\rangle = L_{n,i}^* R_{n,i}$, which sums to one but need not be real or positive. Right-only densities $|R_{n,i}|^2$ are not invariant under similarity transformations and, as we shall see, can be misleading.

2.2 Model

Sites are ordered $A_0, B_0, A_1, \ldots$, and H_{ij} is the amplitude to hop from j to i:

$$H = \sum_n \Big[\big(t_1 + \tfrac{\gamma}{2}\big) c_{A,n}^\dagger c_{B,n} + \big(t_1 - \tfrac{\gamma}{2}\big) c_{B,n}^\dagger c_{A,n} + t_2 \big(c_{A,n+1}^\dagger c_{B,n} + \text{h.c.} \big) \Big]. \tag{3}$$

The intracell hopping is non-reciprocal: $B \to A$ with amplitude $t_1 + \gamma/2$, $A \to B$ with $t_1 - \gamma/2$. The intercell hopping t_2 is reciprocal. All parameters are real.

Bloch Hamiltonian. Fourier transforming over cells,

$$H(k) = \begin{pmatrix} 0 & h_+(k) \\ h_-(k) & 0 \end{pmatrix}, \qquad h_\pm(k) = t_1 \pm \tfrac{\gamma}{2} + t_2 e^{\mp ik}. \tag{4}$$

The eigenvalues satisfy $E^2 = h_+(k) h_-(k)$. Because $h_+ \neq h_-^*$, $E(k)$ is complex in general and traces a loop in the complex plane as k runs over the Brillouin zone. The periodic spectrum is gapless when either factor vanishes for some real k, i.e. when $|t_1 \pm \gamma/2| = t_2$. For $\gamma = 4/3$, $t_2 = 1$ this happens at $t_1 = 1/3$ and $t_1 = 5/3$.

Chiral symmetry. With $\Gamma = \mathrm{diag}(1, -1, 1, -1, \ldots)$, $\Gamma H \Gamma = -H$ still holds, so eigenvalues come in pairs $\pm E$ and zero modes can be chosen on one sublattice.

2.3 Exact similarity map

A diagonal similarity transformation $D = \mathrm{diag}(d_i)$ multiplies element (i, j) of a matrix by d_i/d_j. We look for D such that

$$H_{\mathrm{open}} = D\, H_{\mathrm{SSH}}(v', t_2)\, D^{-1}, \tag{5}$$

where H_{SSH} is a Hermitian SSH chain with reciprocal hoppings v' and t_2.

Intracell bond. Matching the two intracell elements,

$$v' \frac{d_{A_n}}{d_{B_n}} = t_1 + \frac{\gamma}{2}, \qquad v' \frac{d_{B_n}}{d_{A_n}} = t_1 - \frac{\gamma}{2}. \tag{6}$$

Multiplying the two equations eliminates D; dividing them eliminates v':

$$v'^2 = t_1^2 - \frac{\gamma^2}{4}, \qquad s^2 \equiv \left(\frac{d_{A_n}}{d_{B_n}}\right)^2 = \frac{t_1 + \gamma/2}{t_1 - \gamma/2}. \tag{7}$$

Intercell bond. To keep t_2 unchanged we need $d_{A_{n+1}}/d_{B_n} = 1$. Combined with $d_{A_n}/d_{B_n} = s$ this gives the recursion $d_{A_{n+1}} = d_{A_n}/s$, and so

$$d_{A_n} = s^{-n}, \qquad d_{B_n} = s^{-(n+1)}. \tag{8}$$

The A and B sites of one cell need factors differing by s: the map is site-resolved, not cell-resolved.

Consequences.

1. *Spectrum.* Similar matrices have the same eigenvalues, so the open spectrum equals that of $\mathrm{SSH}(v', t_2)$. It is real whenever v' is real, i.e. for $|t_1| > \gamma/2$, despite the non-Hermiticity.
2. *Right eigenvectors.* If $H_{\mathrm{SSH}}\psi = E\psi$, then $R = D\psi$. Its amplitude carries the factor $|s|^{-n}$, so for $|s| > 1$ every right eigenvector piles up on the left. This is the skin effect.
3. *Left eigenvectors.* $H_{\mathrm{open}}^{\dagger} = D^{-\dagger} H_{\mathrm{SSH}} D^{\dagger}$, so $L = D^{-\dagger}\psi$ grows as $|s|^{+n}$ and piles up on the right.
4. *Biorthogonal density.* $L_i^* R_i = (\psi_i^*/d_i)(d_i\psi_i) = |\psi_i|^2$. The biorthogonal density of the open chain is exactly the Hermitian density of the partner chain.

Outside the real region. For $|t_1| < \gamma/2$, v' and s are imaginary. The partner is then complex symmetric rather than Hermitian, and the spectrum is complex. At the exceptional point $t_1 = \gamma/2$ one intracell hopping vanishes, $s = \infty$, the bulk becomes defective with Jordan blocks, and no map exists.

2.4 Edge states and the OBC transition

Because the open chain is similar to $\mathrm{SSH}(v', t_2)$, every spectral result of the SSH chain carries over with $v \to v'$, $w \to t_2$. The edge-state splitting is

$$\min |E| \simeq \frac{t_2^2 - v'^2}{t_2} \left(\frac{v'}{t_2}\right)^N \qquad (|v'| < t_2), \tag{9}$$

and the edge states disappear at $|v'| = t_2$. Using Eq. (7), $|t_1^2 - \gamma^2/4| = t_2^2$, which has two solutions:

$$t_1 = \sqrt{t_2^2 + \gamma^2/4} \qquad (= 1.2019 \text{ at } \gamma = 4/3), \tag{10}$$

$$t_1 = \sqrt{\gamma^2/4 - t_2^2} \qquad (\text{only for } \gamma \geq 2t_2). \tag{11}$$

Neither coincides with the Bloch gap closings $|t_1 \pm \gamma/2| = t_2$. The second transition is where v' is imaginary with $|v'| = t_2$; below it the chain is trivial again.

Parameter-free locator. At $|v'| = t_2$ the partner is a uniform chain of $2N$ sites with hopping t_2, whose eigenvalues are

$$E_m = 2t_2 \cos \frac{m\pi}{2N+1}, \qquad m = 1, \ldots, 2N. \tag{12}$$

The smallest $|E|$ is at $m = N$:

$$\min |E|_{\text{crit}} = 2t_2 \sin \frac{\pi}{2(2N+1)} \;\propto\; \frac{1}{N}. \tag{13}$$

The ratio for two chain lengths is therefore fixed at the transition, with no free parameter:

$$\frac{\min |E|_{30}}{\min |E|_{20}} = \frac{\sin(\pi/122)}{\sin(\pi/82)} \approx \frac{41}{61}. \tag{14}$$

Below the transition the edge splitting decays exponentially and the ratio is smaller; above it $\min |E|$ saturates at the bulk gap and the ratio approaches one. The crossing of this ratio through 41/61 locates the transition.

2.5 Exact zero mode of the odd chain

Take a chain with N A-sites and $N-1$ B-sites, so that it starts and ends on A. A right vector that lives only on A satisfies all A rows of $HR = 0$ trivially. The B row of cell n reads

$$\big(t_1 - \tfrac{\gamma}{2}\big) R_A(n) + t_2\, R_A(n+1) = 0 \quad \Longrightarrow \quad R_A(n) = r_R^n, \qquad r_R = -\frac{t_1 - \gamma/2}{t_2}. \tag{15}$$

The same argument for $H^\dagger$, whose B rows contain the complex conjugate of the $B \to A$ amplitude, gives

$$L_A(n) = r_L^n, \qquad r_L = -\frac{(t_1 + \gamma/2)^*}{t_2^*}. \tag{16}$$

Both are exact zero modes for any N, since with an odd number of sites there is no partner to hybridise with. The biorthogonal density is

$$\langle \Pi_n \rangle = \frac{L_A(n)^* R_A(n)}{\sum_m L_A(m)^* R_A(m)} = \frac{x^n}{\sum_{m=1}^{N} x^m}, \qquad x = r_L^* r_R = \frac{t_1^2 - \gamma^2/4}{t_2^2} = \frac{v'^2}{t_2^2}. \tag{17}$$

$|x| = 1$ is the same condition as $|v'| = t_2$. At $t_1 = 1$, $|r_R| = 1/3$ and $|r_L| = 5/3$: the right vector alone is on the left end and the left vector alone on the right end. Only their product, with $x = 5/9 < 1$, decides that the mode is localised on the left.

2.6 Biorthogonal polarisation and its finite-size term

Kunst *et al.* define the biorthogonal polarisation

$$P = 1 - \lim_{N\to\infty} \frac{1}{N} \Big\langle \psi_L \Big| \sum_n n\, \Pi_n \Big| \psi_R \Big\rangle, \tag{18}$$

which is 1 when the zero mode sits on the left end and 0 when it sits on the right. At finite N it is 1 minus the mean biorthogonal position divided by N:

$$P_N = 1 - \frac{\bar{n}}{N}, \qquad \bar{n} = \frac{\sum_{n=1}^{N} n\, x^n}{\sum_{n=1}^{N} x^n}. \tag{19}$$

Case $|x| < 1$. The sums converge as $N \to \infty$ to $x/(1-x)^2$ and $x/(1-x)$, so $\bar{n} \to 1/(1-x)$ and

$$1 - P_N = \frac{1}{N(1-x)} + O(x^N). \tag{20}$$

Case $|x| > 1$. Substitute $n = N - m$ and $y = 1/x$; then $\bar{n} = N - \sum_m m\, y^m / \sum_m y^m \to N - 1/(x-1)$, so

$$P_N = \frac{1}{N(x-1)} + O(x^{-N}). \tag{21}$$

Case $x = 1$. The density is flat, $\bar{n} = (N+1)/2$ and $P_N = \frac{1}{2} - \frac{1}{2N}$.

Eqs. (20)–(21) hold for negative x as well, which is why P dips slightly below zero on the trivial side for $\gamma = 3$, where v' is imaginary and $x < 0$.

Even chains at $x = -1$. The normalisation $\sum_{n=1}^{N} x^n$ equals $\sum_{n=1}^{N} (-1)^n$ at $x = -1$, which vanishes for even N. The left and right zero modes are then biorthogonally orthogonal, and P is undefined. This happens exactly at the second transition of Eq. (11), and in particular at the parameters $\gamma = 3$ used in Fig. 1 of Ref. [1]. For odd N the norm is -1 and $P = \frac{1}{2} - \frac{1}{2N}$. All polarisation scans therefore use odd N.

2.7 Spectral winding and the origin of the skin effect

For a non-Hermitian Bloch Hamiltonian the periodic spectrum can enclose area in the complex plane. For a reference energy E_b not on the periodic spectrum, define the spectral winding number

$$w(E_b) = \frac{1}{2\pi i} \oint_{\mathrm{BZ}} \frac{d}{dk} \ln \det \big[H(k) - E_b\big]\, dk. \tag{22}$$

Here $\det[H(k) - E_b] = E_b^2 - h_+(k) h_-(k)$, a Laurent polynomial in $z = e^{ik}$, and $w(E_b)$ is the number of its zeros inside the unit circle minus the order of its pole at $z = 0$. For a Hermitian chain the periodic spectrum is a set of real segments with no interior, and $w = 0$ everywhere off the spectrum. Here the periodic loop encloses area, and energies inside it have $w \neq 0$. A nonzero $w(E_b)$ means that $H - E_b$, restricted to a half-infinite chain, has a nontrivial index, and its kernel or the kernel of its adjoint contains a state localised at one end. This is the mechanism of the skin effect [3]: the open-chain eigenvalues lie inside the periodic loop, and every one of them has an eigenvector localised at the end selected by the sign of w.

2.8 Pseudospectra

For a non-normal matrix the eigenvalues alone do not describe how the matrix responds to perturbations. The ϵ-pseudospectrum is the set

$$\sigma_\epsilon(H) = \big\{ z \in \mathbb{C} :\ \|(z - H)^{-1}\| > 1/\epsilon \big\}, \tag{23}$$

equivalently the set of eigenvalues of all matrices $H + \delta H$ with $\|\delta H\| < \epsilon$. For a Hermitian matrix σ_ϵ is the union of discs of radius ϵ around the eigenvalues. For banded Toeplitz matrices with a nonzero spectral winding, the resolvent norm grows exponentially in N for every z inside the periodic loop [7]. The pseudospectrum then fills that region even for tiny ϵ: perturbations at the level of rounding can, in principle, move eigenvalues anywhere inside the loop. This is the worst case; Section 5.6 shows how far the actual errors fall short of it.

2.9 Eigenvalue conditioning

Perturb H by δH. To first order the eigenvalue shift is

$$\delta E_n = \frac{\langle L_n | \delta H | R_n \rangle}{\langle L_n | R_n \rangle} \quad \Longrightarrow \quad |\delta E_n| \le \kappa_n \, \|\delta H\|, \qquad \kappa_n = \frac{\|L_n\| \, \|R_n\|}{|\langle L_n | R_n \rangle|}. \tag{24}$$

For a Hermitian matrix $L_n = R_n$ and $\kappa_n = 1$. Here $R = D\psi$ and $L = D^{-\dagger}\psi$ with $\langle L | R \rangle = \langle \psi | \psi \rangle = 1$, while $\|R\| \, \|L\|$ grows with the spread of D over the chain, $\kappa_n \sim |s|^N$. Rounding in a backward-stable eigensolver acts like $\|\delta H\| \sim \epsilon_{\mathrm{mach}} \|H\|$, so this bound predicts errors of order one once $|s|^N \sim 10^{16}$, i.e. $N \approx 20$ for $t_1 = 0.7$. Section 5.6 shows that the bound is far from tight.

3 Numerical methods

3.1 Exact reference spectrum

For $|t_1| > \gamma/2$ the exact open spectrum is obtained from the Hermitian partner $\mathrm{SSH}(v', t_2)$ with a symmetric eigensolver, which is backward stable with absolute error $O(\epsilon_{\mathrm{mach}}\|H\|)$ and no dependence on non-normality. For $|t_1| < \gamma/2$ the partner is complex symmetric with well-conditioned eigenvectors; it is diagonalised with the general eigensolver and checked against 30-digit arithmetic. Every eigenvalue of the open chain returned by the general eigensolver is compared with this reference.

3.2 General eigensolver and biorthonormalisation

The non-Hermitian open chain is diagonalised by the standard dense algorithm: reduction to upper Hessenberg form by Householder reflections, then implicitly shifted QR iteration to the Schur form $H = QTQ^\dagger$, with eigenvectors from back-substitution in T. Both right and left eigenvectors are requested. Each pair is rescaled so that $\|R_n\| = 1$ and $\langle L_n|R_n\rangle = 1$.

3.3 The QR algorithm and its variants

The general eigensolver proceeds in three stages. First, a diagonal *balancing* similarity is applied to make row and column norms comparable; it changes no eigenvalue and can reduce rounding errors for badly scaled matrices. Second, Householder reflections reduce the matrix to upper Hessenberg form, $H = QH_{\mathrm{hess}}Q^\dagger$, at cost $O(n^3)$. Third, the Hessenberg matrix is driven to quasi-triangular (Schur) form by the implicitly shifted QR iteration: each sweep chases a small bulge down the subdiagonal, and a subdiagonal element is set to zero (deflation) once it is negligible compared with its diagonal neighbours. The eigenvalues are read from the diagonal of the Schur form.

LAPACK chooses between two versions of the third stage by matrix size. Below a crossover (75 in the library used here) it uses the small double-shift QR, which deflates one or two eigenvalues at a time. Above the crossover it uses multishift QR with aggressive early deflation, which computes the Schur form of a trailing window and deflates all eigenvalues whose coupling to the rest of the matrix is below rounding level. Both are backward stable in the sense that the computed eigenvalues are exact for a matrix $H + \delta H$ with $\|\delta H\| = O(\epsilon_{\mathrm{mach}}\|H\|)$; they differ in which δH they produce, and for a matrix whose pseudospectrum is large that difference matters.

3.4 High-precision references

Spot checks are repeated with an arbitrary-precision implementation of the same Hessenberg–QR algorithm at 60 significant digits. At that precision even the pessimistic bound $\kappa_n\epsilon \sim |s|^N \times 10^{-60}$ is below 10^{-27} for the largest case checked ($N = 40$, $|s| = 6.40$), so the result is the exact spectrum to within the printed digits.

3.5 Edge states

- *Even chains.* The two eigenvalues closest to zero form a hybridised pair. The 2×2 matrix $\Gamma_{ij} = \langle L_i|\Gamma|R_j\rangle$ in this pair is diagonalised; its eigenvectors with eigenvalue near ± 1 are the edge states on A and on B. They are then re-biorthonormalised.
- *Odd chains.* The zero mode is selected as the eigenvector with least weight on B (exactly zero for the true mode), not the one with the smallest $|E|$, because rounding can place a bulk eigenvalue closer to zero.

3.6 Skin profile, transitions and polarisation

- *Skin decay.* The bulk eigenvectors ($|E| > 10^{-3}$) are normalised and their cell weights $\sum_\alpha |R_{\alpha,n}|^2$ summed, which averages out the standing-wave oscillations of the individual states. A straight line is fitted to the logarithm of the summed weight, excluding two cells at each end; the exponential of half the slope is the amplitude ratio per cell, compared with $1/|s|$.
- *Transition.* $\min|E|_{30}/\min|E|_{20}$ is computed on a fine grid in t_1, and its crossing through 41/61 is found by linear interpolation between grid points.
- *Phase diagram.* $\log_{10}\min|E|$ of a 30-cell chain on a 201×151 grid in (t_1, γ).
- *Polarisation.* P is computed both from the closed form and from the numerical biorthogonal zero mode of the odd chain, $P_N = 1 - \langle L|\hat{n}|R\rangle/N$.

3.7 Accuracy diagnostics

The error of the general eigensolver is the largest distance from any computed eigenvalue to the nearest exact eigenvalue. Non-normality is measured by the 2-norm condition number of the matrix of unit right eigenvectors, $\mathrm{cond}(R) = \sigma_{\max}/\sigma_{\min}$, from its singular values. Spot checks are repeated in 60-digit arithmetic, where rounding is irrelevant at these sizes. A Hermitian chain ($\gamma = 0$) of the same size is used as a control.

4 Verification

The tests are built around the exact similarity map: almost every numerical statement has an independent exact counterpart.

1. Structure of the model. The open-chain matrix elements must be those of the Hamiltonian above; the odd chain must have an odd number of sites; for $\gamma = 0$ the model must reduce to the Hermitian SSH chain; the spectrum must come in $\pm E$ pairs; and the Bloch Hamiltonian must equal the Fourier transform of the periodic ring, with gap closings at $|t_1 \pm \gamma/2| = t_2$.

2. The similarity map. $DH_{\mathrm{SSH}}D^{-1}$ must reproduce H_{open} element by element, to round-off. As a negative control, the per-cell version of D (one factor per cell) must fail this test. The open spectrum must then equal that of $\mathrm{SSH}(v', t_2)$, and, in the imaginary-v' region, that of the complex-symmetric partner computed in high precision.

3. Eigenvectors. Left and right eigenvectors must be biorthonormal. The right eigenvectors must equal $cD\psi$ for the partner eigenvectors ψ, the decay per cell must approach $1/|s|$, and for $\gamma = 0$ there must be no skin effect. The biorthogonal edge density must equal the Hermitian edge density of the partner, and the right and left edge vectors must sit at opposite ends.

4. The transition. Below the OBC transition $\min|E|$ must decay exponentially in N with slope $\ln(v'/t_2)$; above it, it must saturate. The finite-size crossing must follow $\sqrt{t_2^2 + \gamma^2/4}$ for several γ, and, as a negative control, nothing may happen at the Bloch closings.

5. Polarisation. The factors r_R, r_L must match the paper; the odd-chain zero mode must be exact and supported on A; the biorthogonal density must match $x^n/\sum x^m$; the numerical P must match the closed form, and must approach quantisation with the $1/N$ forms of Eqs. (20)–(21). P must jump at the OBC point, not at the Bloch point; the second transition must appear for large γ; even N must be singular at $x = -1$; and the Hermitian limit must give $P = 1$.

6. Eigensolver accuracy. Below 76 sites the general eigensolver must agree with the exact spectrum; above, it must be observed to fail. Because this boundary is set inside the linear-algebra library, the platform and library versions are recorded with the results.

5 Results

5.1 Periodic versus open spectrum

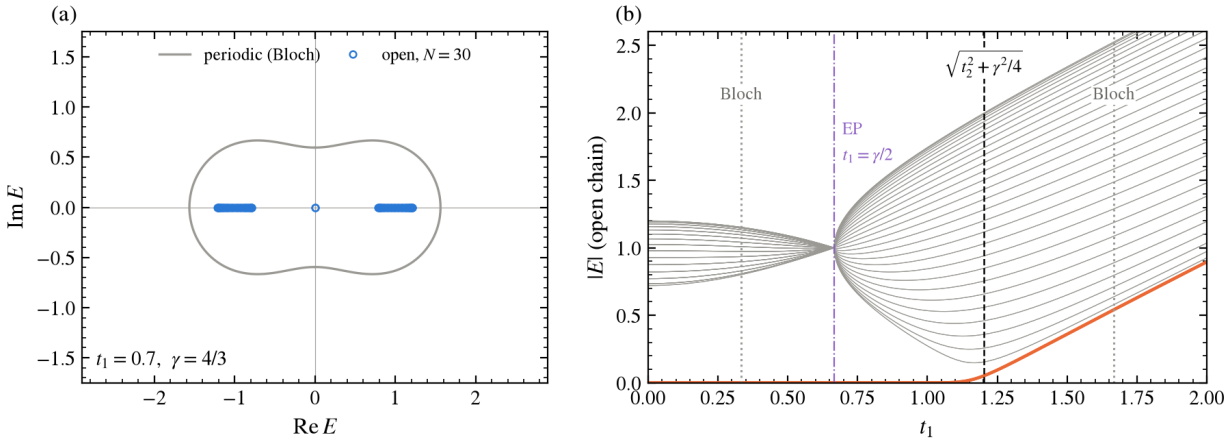


Figure 1: (a) Complex spectrum at $t_1 = 0.7$: periodic (grey loop, $|\operatorname{Im} E| \le 0.667$) vs open $N = 30$ (blue, real, zero-mode pair at the origin). (b) $|E|$ of a 30-cell open chain vs t_1: the near-zero pair leaves $E = 0$ at the OBC transition $t_1 = 1.2019$ (dashed); nothing happens at the Bloch closings $1/3, 5/3$ (dotted).

5.2 Skin effect

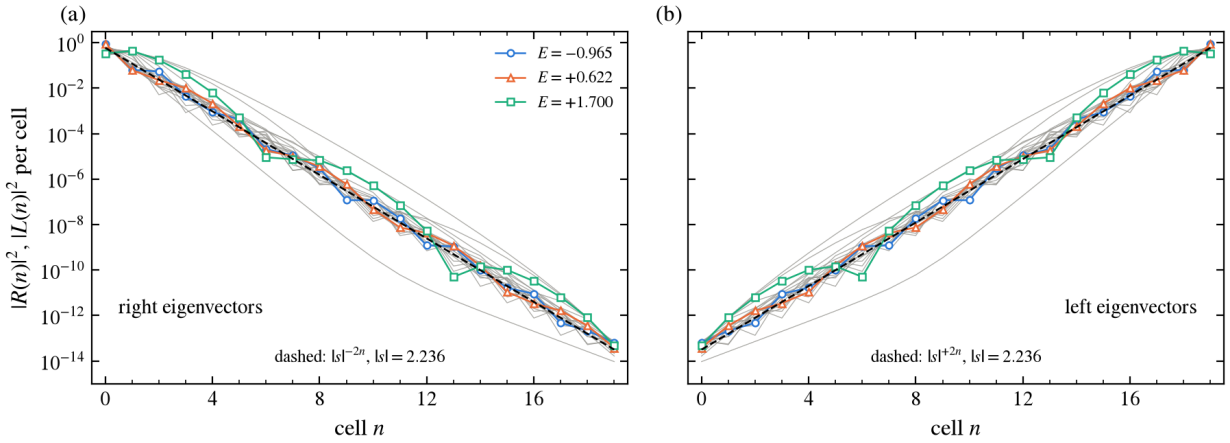


Figure 2: Weight per cell of all 40 eigenvectors of a 20-cell chain at $t_1 = 1.0$. (a) Right eigenvectors pile up on the left, (b) left eigenvectors on the right; dashed: $|s|^{\mp 2n}$, $|s| = \sqrt{5}$.

t_1	$1/\lvert s\rvert$	right decay/cell ($N = 12$)	rel. err.	left growth/cell	$\lvert s\rvert$
0.7	0.1562	0.1591	+1.9%	6.287	6.403
1.0	0.4472	0.4508	+0.8%	2.218	2.236
1.4	0.5957	0.5959	+0.0%	1.678	1.679

The residual is the boundary shape of the standing waves and halves from $N = 12$ to 20. The exact statement $R_n/\psi_n = cD_n$ holds to 10^{-8}. Note that $0.159 \approx 1/s$ is the *bulk* decay; the zero mode decays faster, by $|r_R| = 0.033$ per cell at $t_1 = 0.7$.

5.3 Open chain against the exact solution

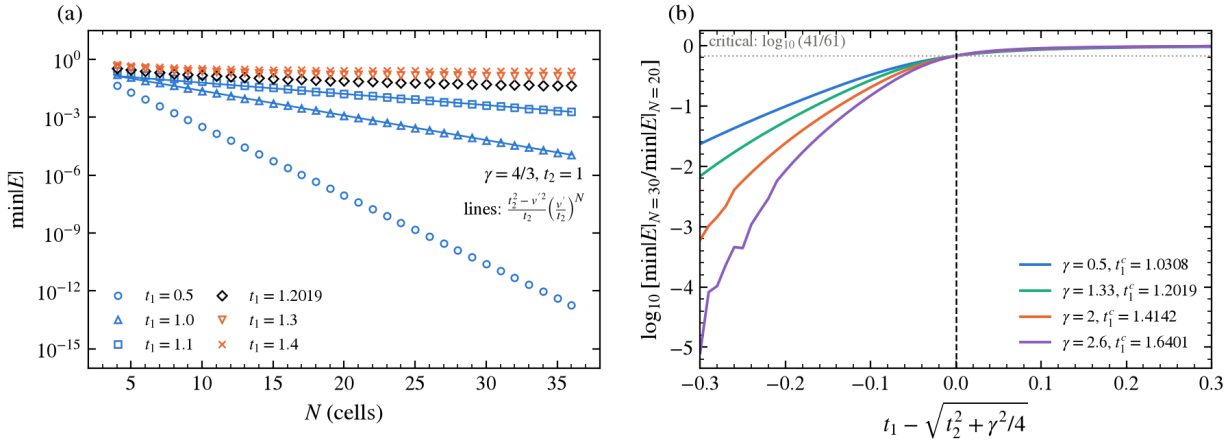


Figure 3: (a) $\min|E|$ vs N: exponential decay below the transition (lines: SSH splitting with $v \to v'$), $1/N$ at it, saturation above. (b) $\min|E|_{30}/\min|E|_{20}$ vs distance from $\sqrt{t_2^2 + \gamma^2/4}$ for four γ: all cross $\log_{10}(41/61)$ at zero.

t_1	$N = 10$	$N = 20$	$N = 30$	
0.5	3.3×10^{-4}	9.2×10^{-8}	2.6×10^{-11}	edge states (v' imaginary)
1.0	2.38×10^{-2}	1.24×10^{-3}	6.59×10^{-5}	edge states
1.1	7.3×10^{-2}	1.7×10^{-2}	4.3×10^{-3}	edge states, $\xi = 7.5$ cells
1.3	0.239	0.167	0.145	no edge states
1.4	0.337	0.270	0.252	no edge states

5.4 Phase diagram

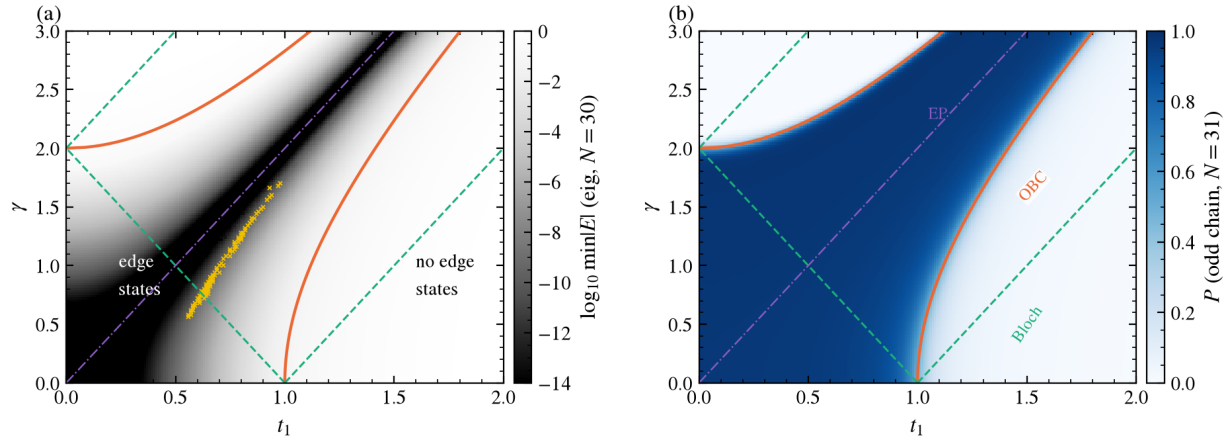


Figure 4: (a) $\log_{10}\min|E|$ of a 30-cell chain from `eig` (201×151 grid). Orange: OBC transitions $|t_1^2 - \gamma^2/4| = t_2^2$; green dashed: Bloch closings; purple: exceptional line; yellow crosses: points where `eig`'s $\min|E|$ is off by $> 2\times$. (b) Biorthogonal polarisation of the odd chain, $N = 31$.

γ	$P = 1/2$ at t_1	analytic
0.50	1.028	1.031
1.34	1.201	1.204
2.00	1.412	1.414
2.60	0.832, 1.638	0.831, 1.640

The offset is the $1/N$ term, $P_N = 1/2 - 1/(2N)$ at the transition.

Even at $N = 30$ there is a trap. The largest eigenvalue error on the grid is only 10^{-8}, yet at 90 of 26 308 points `eig` returns $\min|E|$ more than $2\times$ wrong (worst: 10^{-17} vs exact 2×10^{-10}). An absolute error bound says nothing about the relative accuracy of a tiny eigenvalue; without the exact solution, these points would look like especially good edge states.

5.5 Densities and polarisation

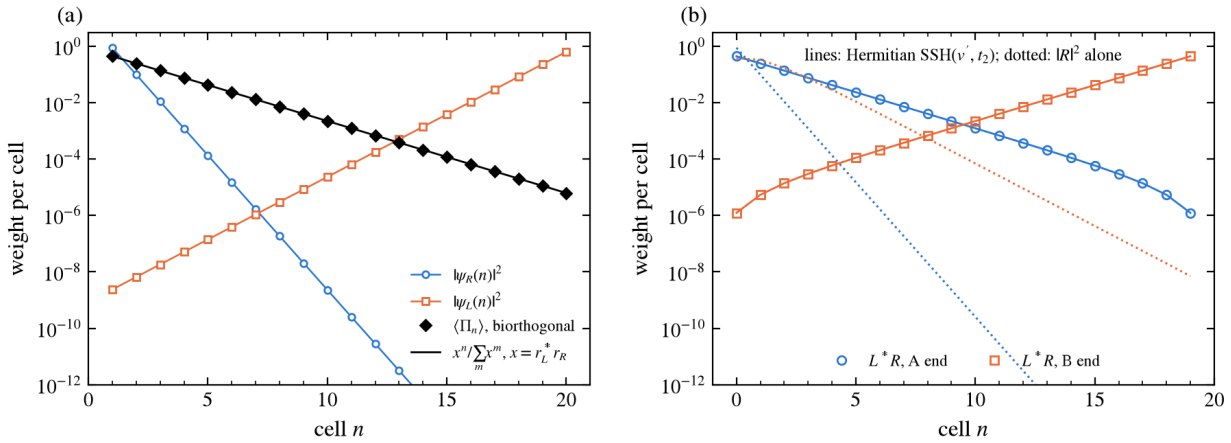


Figure 5: (a) Exact zero mode of the odd chain ($N = 20$, $t_1 = 1$): $|\psi_R|^2$ decays from the left ($r_R = -1/3$), $|\psi_L|^2$ grows to the right ($r_L = -5/3$); the biorthogonal density decays as x^n, $x = 0.556$. (b) Even chain: L^*R (symbols) on the Hermitian SSH(v', t_2) densities (lines), difference $\leq 2.8 \times 10^{-12}$; dotted: $|R|^2$ alone, misleading.

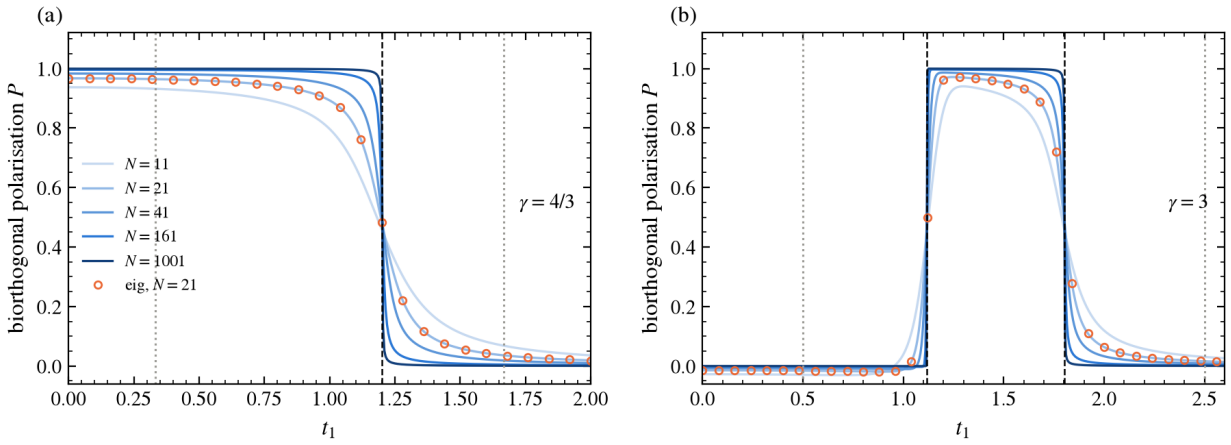


Figure 6: P vs t_1: closed form for $N = 11$ to 1001 (lines), `eig` at $N = 21$ (circles). (a) $\gamma = 4/3$. (b) $\gamma = 3$ (paper's Fig. 1), transitions at $t_1 = 1.118$ and 1.803.

γ	N	P at OBC point(s)	P at Bloch points
4/3	21	0.476	0.964, 0.036
4/3	1001	0.4995	0.9993, 0.0007
3	21	0.476, 0.476	−0.016, 0.016
3	1001	0.4995, 0.4995	−0.0003, 0.0003

P steps at the OBC transitions and ignores the Bloch points. At $\gamma = 3$, `eig` agrees with the closed form only to 3.5×10^{-5} because the norm $\sum x^n$ nearly cancels near $x = -1$.

5.6 When does `eig` fail?

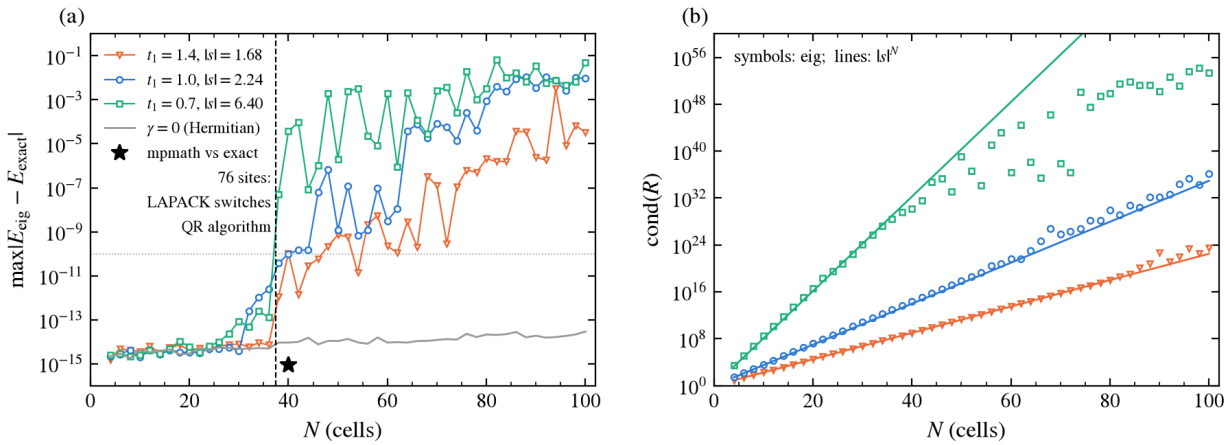


Figure 7: (a) Largest distance from an `eig` eigenvalue to the exact spectrum vs N; grey: Hermitian control $\gamma = 0$; star: 60-digit arithmetic. (b) cond(R) (symbols) and $|s|^N$ (lines).

t_1	$\lvert s\rvert$	$N = 30$	$N = 36$	$N = 38$	$N = 40$	$N = 60$	$N = 100$
1.4	1.68	8.0×10^{-15}	7.1×10^{-15}	1.1×10^{-12}	1.0×10^{-10}	2.2×10^{-10}	3.4×10^{-5}
1.0	2.24	3.8×10^{-15}	2.4×10^{-12}	3.7×10^{-11}	1.0×10^{-10}	3.2×10^{-9}	9.4×10^{-3}
0.7	6.40	8.7×10^{-14}	1.3×10^{-13}	5.0×10^{-8}	3.7×10^{-5}	2.0×10^{-3}	4.9×10^{-2}

cond(R) follows $|s|^N$ up to 10^{24} at $N = 30$, yet the eigenvalues stay accurate to 10^{-13}. All curves then jump together between 74 and 76 sites. LAPACK's `dhseqr` uses the small double-shift QR (`dlahqr`) up to $n = 75$ (`NMIN` in `iparmq`) and multishift QR with aggressive early deflation [8] above; the latter breaks on these graded matrices. A Schur decomposition without balancing is still accurate to 3×10^{-11} below the switch, so balancing is not the explanation. The Hermitian control stays below 3×10^{-14} to $N = 100$.

With NumPy 2.4.6 and OpenBLAS 0.3.31, `eig` errs by more than 10^{-10} from $N = 38$ ($t_1 = 0.7$), 40 ($t_1 = 1.4$) and 42 ($t_1 = 1.0$). At $N = 40$, $t_1 = 0.7$, 60-digit arithmetic agrees with the exact formula to 10^{-15} while `eig` is off by 3.7×10^{-5}. The size at which this happens is a LAPACK tuning parameter, so a different build can move it.

6 Discussion

Why the Bloch winding fails. Bloch states are plane waves, $|e^{ikn}| = 1$; the open-chain eigenstates grow or decay as $|s|^{\mp n}$. The periodic spectrum belongs to a different operator. The open chain is similar to SSH(v', t_2), whose winding changes at $|v'| = t_2$; equivalently, it lives on the generalised Brillouin zone $|e^{ik}| = 1/|s|$ of Yao and Wang.

What the similarity map does. It leaves the spectrum unchanged and multiplies every eigenvector by the same exponential profile. Observables built from L^*R (densities, P) do not see D; observables from $|R|^2$ alone carry the full skin profile.

When not to trust `eig`. When cond(R) $\gg 1$. But the condition-number bound is too pessimistic below 76 sites, something else entirely sets the failure above, and small absolute errors can make tiny eigenvalues wrong by orders of magnitude. Remedies: an exact reference, high-precision spot checks, checks against N and algorithm, and staying below the size where they fail.

Limitations and open questions. (i) Why `dlahqr` stays accurate at $|s|^N = 10^{48}$ is not pinned down. (ii) With disorder the map survives with site-dependent s_n, v'_n, suggesting a transition at $\langle \ln |v'_n| \rangle = \ln t_2$; not yet tested. (iii) The crossover scale $\alpha \approx 0.58$ quoted in the paper is not explained by $\ln |s| = 0.80$. (iv) Whether $L^* R$ is directly measurable in photonic or circuit lattices remains open.

7 Conclusion

An exact similarity map turns a numerically treacherous non-Hermitian problem into a Hermitian one with a known answer. Using it as reference, we reproduced the biorthogonal bulk–boundary correspondence, located the OBC transitions to four digits with a parameter-free method, derived the finite-size form of the biorthogonal polarisation and an even-chain singularity, and showed that the failure of standard eigensolvers on skin-effect matrices is set by an algorithm switch inside LAPACK, not by the eigenvector condition number.

References

[1] F. K. Kunst, E. Edvardsson, J. C. Budich, E. J. Bergholtz, *Biorthogonal bulk-boundary correspondence in non-Hermitian systems*, Phys. Rev. Lett. **121**, 026808 (2018).

[2] S. Yao, Z. Wang, *Edge states and topological invariants of non-Hermitian systems*, Phys. Rev. Lett. **121**, 086803 (2018).

[3] E. J. Bergholtz, J. C. Budich, F. K. Kunst, *Exceptional topology of non-Hermitian systems*, Rev. Mod. Phys. **93**, 015005 (2021).

[4] D. C. Brody, *Biorthogonal quantum mechanics*, J. Phys. A **47**, 035305 (2014).

[5] T. E. Lee, *Anomalous edge state in a non-Hermitian lattice*, Phys. Rev. Lett. **116**, 133903 (2016).

[6] V. M. Martinez Alvarez, J. E. Barrios Vargas, L. E. F. Foa Torres, *Non-Hermitian robust edge states in one dimension*, Phys. Rev. B **97**, 121401(R) (2018).

[7] L. N. Trefethen, M. Embree, *Spectra and Pseudospectra*, Princeton University Press (2005).

[8] K. Braman, R. Byers, R. Mathias, *The multishift QR algorithm. Part II: Aggressive early deflation*, SIAM J. Matrix Anal. Appl. **23**, 948 (2002).